**Learning Objective 2.7:** I can measure the radiation pattern of an antenna and extract gain, beamwidth, sidelobe level, and polarization from the data.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Is the range honest?** You are setting up a pattern range at $f = 2.45$ GHz. The antenna under test (AUT) is a pyramidal horn with a $24 \times 17$ cm aperture. The lab bench gives you a separation of $r = 3.0$ m.

   (a) Find the wavelength and the AUT's largest dimension $D$, then evaluate the far-field criterion $r \geqslant 2D^2/\lambda$.

   $2D^2/\lambda = \boxed{1.41 \text{ m}}$

   The largest dimension of a rectangular aperture is its diagonal, not a side.

   $$\lambda = \frac{3 \times 10^8}{2.45 \times 10^9} = 0.1224 \text{ m} = 12.2 \text{ cm}$$
   $$D = \sqrt{(0.24)^2 + (0.17)^2} = 0.294 \text{ m}$$
   $$\frac{2D^2}{\lambda} = \frac{2(0.294)^2}{0.1224} = 1.41 \text{ m}$$

   (b) The far field also requires $r \geqslant 5D$ and $r \geqslant 10\lambda$. Evaluate both, state which criterion binds, and say whether 3.0 m is acceptable.

   binding $r = \boxed{1.47 \text{ m}}$

   $$5D = 5(0.294) = 1.47 \text{ m}$$
   $$10\lambda = 10(0.1224) = 1.22 \text{ m}$$

   The binding criterion is $5D = 1.47$ m — for a physically small antenna the famous $2D^2/\lambda$ rule is often *not* the one that decides. A 3.0 m range clears the binding value by a factor of 2.0, so it is comfortably acceptable.

   (c) In one or two sentences: if you ran this measurement at $r = 0.5$ m instead, what specifically would go wrong with the recorded pattern, and would the measured HPBW read too wide or too narrow?

   At 0.5 m the spherical wave from the source is not flat across the AUT aperture, so the edges of the aperture are excited with the wrong relative phase. That phase error broadens the main beam and fills in the nulls, and the sidelobes smear into the beam skirt. The measured HPBW reads **too wide**, and the sidelobe structure you report would not be the antenna's far-field pattern at all.

(d) Your calibrated reference horn has a $34 \times 25$ cm aperture. Recompute the far-field distance for *it*, and state what that means for the range you just declared acceptable.

$2D^2/\lambda = \boxed{2.91 \text{ m}}$

$$D = \sqrt{(0.34)^2 + (0.25)^2} = 0.422 \text{ m}$$
$$\frac{2D^2}{\lambda} = \frac{2(0.422)^2}{0.1224} = 2.91 \text{ m} \qquad (5D = 2.11 \text{ m})$$

The *reference* sizes the range, not the AUT, and 3.0 m clears 2.91 m by only 3%. The gain comparison is therefore the one measurement standing at the edge of the far field; say so in the report, or lengthen the range to roughly 5.8 m for a factor-of-two margin.

2. **Reducing a measured cut.** The table below is one measured E-plane cut of an aperture antenna, in dBm at the receiver. The cut was symmetric about $0°$ to within 0.3 dB, so only $\theta \geqslant 0$ is tabulated. With the source switched off and every other setting unchanged, the receiver read $-78.0$ dBm.

| $\theta$ | level | $\theta$ | level | $\theta$ | level |
|---|---|---|---|---|---|
| $0°$ | $-35.6$ | $18°$ | $-49.5$ | $48°$ | $-57.5$ |
| $2°$ | $-36.0$ | $20°$ | $-48.7$ | $60°$ | $-59.8$ |
| $4°$ | $-36.7$ | $22°$ | $-50.0$ | $75°$ | $-62.5$ |
| $6°$ | $-38.3$ | $24°$ | $-52.7$ | $90°$ | $-64.0$ |
| $8°$ | $-41.4$ | $28°$ | $-77.6$ | $120°$ | $-66.5$ |
| $10°$ | $-45.4$ | $32°$ | $-55.6$ | $150°$ | $-63.0$ |
| $12°$ | $-53.5$ | $36°$ | $-53.3$ | $180°$ | $-60.6$ |
| $14°$ | $-66.4$ | $40°$ | $-56.7$ | | |
| $16°$ | $-52.6$ | $44°$ | $-70.6$ | | |

(a) Extract the half-power beamwidth. Interpolate between samples — do not snap to the nearest grid point.

$\theta_{HP} =$ $\boxed{12.4°}$

The peak is $-35.6$ dBm, so the half-power level is $-38.6$ dBm. That falls between $6°$ ($-38.3$ dBm) and $8°$ ($-41.4$ dBm):

$$\theta_{-3} = 6 + 2\left(\frac{38.6 - 38.3}{41.4 - 38.3}\right) = 6 + 2(0.097) = 6.19°$$

$$\theta_{HP} = 2(6.19) = 12.4°$$

(b) Find the first sidelobe level relative to the peak, and compare it with the canonical value for a uniformly illuminated aperture.

SLL = $\boxed{-13.1 \text{ dB}}$

The first null is near $14°$; the next lobe peaks at $20°$ with $-48.7$ dBm:

$$\text{SLL} = -48.7 - (-35.6) = -13.1 \text{ dB}$$

A uniformly illuminated aperture gives $-13.3$ dB, so this aperture is essentially uniformly illuminated — no taper. The second lobe at $36°$ gives $-53.3 + 35.6 = -17.7$ dB, against the uniform-aperture value of $-17.9$ dB: the same story.

(c) Compute the front-to-back ratio.

$$\text{F/B} = -35.6 - (-60.6) = 25.0 \text{ dB}$$

F/B = $\boxed{25.0 \text{ dB}}$

(d) The operator used a $2°$ step inside the main beam. Using the rule of thumb for angle step, was that fine, wasteful, or too coarse? Answer in one sentence, with the number that justifies it.

The rule is step $\leqslant \theta_{HP}/5 = 12.4/5 = 2.5°$, so a $2°$ step satisfies it with a little margin — fine, not wasteful. (The $4°$ and coarser steps used past $24°$ do *not* resolve the sidelobe peaks well, which is why the extracted sidelobe levels deserve a wider error bar than the beamwidth does.)

3. **Gain by comparison, and polarization.** On the range of Question 1 you record a peak of −35.6 dBm with the AUT installed. You then swap in the calibrated reference horn ($G_{ref}$ = 15.0 dBi ), change nothing else, and record −32.4 dBm .

   (a) Find the gain of the AUT in dBi.

   In dB the comparison method is a single subtraction, because transmit power, path loss, cable loss and source gain are common to both measurements and cancel:

   $G_{AUT}$ = | 11.8 dBi |

$$\begin{aligned} G_{AUT} &= G_{ref} + (P_{AUT} - P_{ref}) \\ &= 15.0 + (-35.6 + 32.4) \\ &= 15.0 - 3.2 = 11.8 \text{ dBi} \end{aligned}$$

   (b) The procedure insists that you change *nothing* between the two measurements. Name two things that would silently corrupt the result if you changed them, and say what the resulting gain error would look like.

   Any change that is not common to both measurements survives the subtraction and becomes gain error. Examples: moving or reseating a cable (a fraction of a dB of extra loss appears as a fixed gain offset); re-peaking sloppily so the reference is measured off boresight (the reference reads low, so the AUT gain reads *high*); changing the transmit level or the resolution bandwidth between the two runs; moving either mast so the range length differs. All of these are biases, not noise — repeating the sweep will not reveal them.

   (c) You rotate the source antenna 90° about the range axis and re-peak: the boresight level drops to −59.9 dBm . Compute the cross-polarization discrimination.

   XPD = | 24.3 dB |

$$\text{XPD} = P_{co} - P_{cross} = -35.6 - (-59.9) = 24.3 \text{ dB}$$

   That is squarely in the healthy 20–30 dB band for a linearly polarized antenna.

   (d) Suppose instead the cross-pol peak had landed only 6.0 dB above your noise floor. By how much would the measured cross-pol level overstate the true one, and what does that do to your reported XPD?

   error = | 1.25 dB |

   The reading is signal plus noise. If measured is 6.0 dB above the floor, then measured/noise = $10^{0.6}$ = 3.98, so signal/noise = 2.98 and

$$\text{error} = 10 \log_{10}\left(\frac{3.98}{2.98}\right) = 1.25 \text{ dB}$$

   The cross-pol level reads 1.25 dB high, so the true XPD is 1.25 dB *better* than reported. XPD measured near the floor is therefore a **lower bound**: report it as "XPD ⩾" with the floor quoted beside it.

4. **What the noise floor is doing to your numbers.** A measured pattern reaches its peak at $-35.6$ dBm ; with the source off, the receiver reads $-78.0$ dBm . Levels below are quoted relative to the peak.

   (a) A null in a different data set measures $-25.0$ dB on a system whose floor sits at $-30.0$ dB relative to the peak. Estimate the true depth of that null.

   true null = $\boxed{-26.7 \text{ dB}}$

   Signal and noise add in *power*, so subtract them in power:

$$\begin{aligned} p_{true} &= 10^{-2.50} - 10^{-3.00} \\ &= 3.162 \times 10^{-3} - 1.000 \times 10^{-3} = 2.162 \times 10^{-3} \\ \text{null} &= 10 \log_{10} \left(2.162 \times 10^{-3}\right) = -26.7 \text{ dB} \end{aligned}$$

   The null is 1.7 dB deeper than it measured.

   (b) Now do the same subtraction for a reading of $-42.0$ dB on this system, whose floor is at $-78.0 + 35.6 = -42.4$ dB relative. Comment on the answer you get.

   true null = $\boxed{\leqslant -42 \text{ dB}}$

$$\begin{aligned} p_{true} &= 10^{-4.20} - 10^{-4.24} \\ &= 6.310 \times 10^{-5} - 5.754 \times 10^{-5} = 5.55 \times 10^{-6} \\ \text{null} &= 10 \log_{10} \left(5.55 \times 10^{-6}\right) = -52.6 \text{ dB} \end{aligned}$$

   The number is worthless. It is the difference of two nearly equal quantities: a 0.3 dB error in either reading moves the answer by tens of dB. The only defensible statement is "the null is below $-42$ dB , floor-limited."

   (c) You average 16 sweeps instead of 1. In one or two sentences, say what improves and what does not, and why.

   Averaging power measurements incoherently reduces the *variance* of the noise — the fuzz on the trace shrinks by about $\sqrt{16} = 4$ — but the *mean* noise power is unchanged, so the floor does not move. You get a prettier trace with exactly the same dynamic range. Lowering the floor requires more transmit power, a narrower resolution bandwidth, or a quieter receiver.

   (d) State this system's dynamic range, then rank HPBW (12.4°), first sidelobe ($-13.1$ dB ) and null depth ($-42.0$ dB ) by how much you trust them, with the margin above the floor for each.

   DR = $\boxed{42.4 \text{ dB}}$

$$\text{DR} = -35.6 - (-78.0) = 42.4 \text{ dB}$$

   HPBW is read 3 dB down, roughly 39 dB above the floor — trust it first. The first sidelobe sits 29 dB above the floor, so the noise adds about 0.005 dB to it — trust it too. The null is *at* the floor, margin $\approx 0$ dB — report it as a bound, never as a value.

5. **The uncertainty paragraph you have to write.** Your report claims $G_{AUT} = 11.8$ dBi for the horn of Question 3, against an aperture-formula prediction of 12.3 dBi (using $\eta_{\text{ap}} = 0.5$).

   (a) Your peak-find left the AUT 1.5° off boresight on a 12.4° beam. Using a parabolic approximation to the main beam near its peak, how much does the recorded peak read low?

   $\Delta = \boxed{-0.18 \text{ dB}}$

   Near the peak the pattern in dB is parabolic and passes through −3 dB at half the beamwidth, 6.2°:

   $$\Delta = -3\left(\frac{1.5}{6.2}\right)^2 = -3(0.0585)$$
   $$= -0.18 \text{ dB}$$

   Small — but it is a **bias**, always in the same direction, so repeating the sweep will never remove it.

   (b) A floor reflection arrives 20 dB below the direct path. Find the peak-to-peak ripple it puts on your pattern, then find how far down the reflection would have to be to hold the ripple inside ±0.2 dB .

   ripple $= \boxed{1.7 \text{ dB p-p}}$

   Voltage ratio $10^{-20/20} = 0.10$; the two paths add and subtract:

   $$\max = 20\log_{10}(1.10) = +0.83 \text{ dB}$$
   $$\min = 20\log_{10}(0.90) = -0.92 \text{ dB}$$

   so about 1.7 dB peak to peak. For ±0.2 dB we need $20\log_{10}(1+x) = 0.2$, giving $x = 0.023$, i.e. the reflection must be $20\log_{10}(0.023) = -32.6$ dB below the direct path. That is what the absorber on the bounce point is buying you.

   (c) Combine the error sources — reference tolerance 0.5 dB , ripple 0.9 dB , alignment 0.2 dB , sweep-to-sweep repeatability 0.3 dB — into one uncertainty, then write the two-sentence verdict on the 0.5 dB gap between your measurement and the prediction.

   Independent contributions add in quadrature:

   $$u = \sqrt{0.5^2 + 0.9^2 + 0.2^2 + 0.3^2} = \sqrt{1.19} = 1.1 \text{ dB}$$

   Verdict: the measured 11.8 dBi and the predicted 12.3 dBi differ by 0.5 dB , well inside the ±1.1 dB uncertainty, so the measurement **agrees** with prediction and no physical explanation is required. Note also that the prediction itself is soft — it assumed $\eta_{\text{ap}} = 0.5$, and real horns run anywhere from 0.4 to 0.6, which is a further ±1 dB on the number being compared against.

**Documentation:** ______________________________